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Technical Achievements for Successful Development of Sour Gas Fields in Sichuan Basin: An overview of Series Cost-Effective and Economically Efficient China Solutions

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Abstract

Over the past decade, several sour gas fields including Puguang, Yuanba, and LuoJiazhai characterized by varying reservoir properties and high H₂S and CO₂ concentrations have been successfully and economical developed in the Sichuan Basin, making China the fourth largest sour gas production country in the world, following Russia, Kazakhstan, and Qatar. In 2024, nearly 20 billion cubic meters of sour gas were produced in Sichuan Basin, along with more than 2.2 million tons of sulfur products, which achieved giant economic and social benefits. These achievements are inseparable from a series of technical research efforts aimed at overcoming the challenges associated with the cost-effective and economically efficient development of sour gas reservoirs.

In this study, the key technologies implicated and lessons learned during the process of sour gas development in Sichuan basin are reviewed, elucidated, and evaluated. Specifically, sour gas development full chain technologies including reservoir description, production simulation with sulfur deposition, safety and faster drilling and completion, sulfur deposition management, water invasion management, corrosion protection, monitoring, and detection, sour gas purification, sulfur offtake and uses, and 100% HSE strategies formed the cost-effective China solutions. Accompanying the practice of sour gas development

in the last ten years, several key insights have been gained through the application of the aforementioned technologies.

Appropriate management of H₂S exposure and corrosion of equipment and facilities is the primary consideration. As for the geology and reservoir engineering of sour gas, gas-liquid-solid production simulation model and water invasion control strategies are proposed, while the deposition of element sulfur subsurface and water invasion play negative effects of the recovery factor of sour gas. Meanwhile, in the drilling and completion phase, sulfur-resistant tools and equipment not only determined the well construction timeline but also significantly impacted the overall cost of sour gas development, all of which are domestically designed and manufactured in China. Regarding the purification of sour gas and sulfur recovery, the gathering pipelines, purification facilities, and desulfurization catalysts are all domestically developed technologies in China, while the price of sulfur plays an important role in the methodology of processing sulfur chemicals. Last but not least, with those cost-effective and economically efficient equipment and techniques, the economics of sour gas development has been significantly improved.

This research summarizes the sour gas development technologies, experiences, and lessons learned over the past decade in the Sichuan Basin. To the best of our knowledge, this is the first systematic overview of the technical achievements for sour gas development within Sichuan Basin. It is essential as a reference for the developing of similar sour gas fields with those cost-effective and economically efficient China solutions, especially for sour gas fields in the Middle East area.

Keywords: Sour Gas, Technical Achievements, China Solutions, Cost-Effective and Economically Efficient, Sichuan Basin

Introduction

Globally, sour gas reservoirs, characterized by significant hydrogen sulfide (H₂S) content, constitute a substantial portion of natural gas resources, estimated at approximately 40% of total reserves (Liu et al., 2012). These reservoirs, often located in high-pressure, high-temperature (HPHT) formations and containing high levels of both H₂S and carbon dioxide (CO₂), present a formidable array of engineering and safety challenges spanning the entire development lifecycle (Hu et al., 2024). The toxic and corrosive nature of sour gas necessitates specialized technologies and stringent operational protocols. Nevertheless, the successful exploitation of historic fields such as Canada's Caroline, France's Lacq, and Russia's Astrakhan established the foundational feasibility of sour gas production.

In the contemporary energy landscape, which is increasingly focused on cleaner fuels (Wang et al., 2023), sour gas is gaining prominence as a critical transitional resource. China's Sichuan Basin has become a global focal point for these efforts, positioning the nation among the world's foremost sour gas producers. Over the last decade, Chinese operators have brought several world-class, high-sulfur gas fields online, including Puguang fields, Yuanba fields and Chuandongbei cluster (Luojiashai, Tieshanpo, and Dukouhe-Qilibei) (Wen et al., 2024). These developments have been propelled by significant, targeted technological innovation. By 2024, this concerted effort resulted in continued production growth from the region's primary operators. One oil company's domestic natural gas output grew by 4.6% to 140.36 bcm, with subsidiary seeing a 6.4% increase and achieving an oil and gas equivalent output exceeding 35 million tons. Concurrently, another company's natural gas production rose by 4.7% to approximately 39.6 bcm, with surpassing 10 bcm in annual production for the first time. These figures, a significant portion of which is sour gas, are accompanied by the recovery of millions of tons of elemental sulfur as a marketable byproduct, turning a hazardous contaminant into a valuable commodity (Ge et al., 2023).

This progress has been punctuated by recent landmark achievements that underscore China's maturing capabilities. In May 2023, the Tieshanpo (TSP) field was commissioned, which contains H₂S concentrations up to 15.54%. This was followed by the start of production at the Dukouhe-Qilibei (DKH-QLB) field

on June, 2025, which features an even higher H₂S content of up to 17.06%, making it the most sulfur-rich integrated gas field currently in production in China. The successful, independent development of these fields, which were part of a block previously relinquished by international partner Chevron, signifies a crucial strategic evolution. It demonstrates a transition from a period of technology acquisition through partnership to one of self-reliant technological leadership, where domestic entities now possess the capability and risk tolerance to tackle projects at the frontier of operational difficulty. Further cementing this trend, one company brought its West Sichuan Gas Field online in March 2024, a project designed for an annual output of 2 bcm of gas and 130,000 tons of sulfur, featuring a state-of-the-art closed-loop system with over 99.9% sulfur recovery and zero liquid discharge.

Crucially, these accomplishments were enabled by a comprehensive set of engineering solutions tailored to the unique challenges of sour gas. Within a decade, Chinese operators and researchers developed an end-to-end technology and management system for high-sulfur gas fields, encompassing:

1. Advanced reservoir characterization and flow modeling that account for sulfur precipitation and water encroachment.
2. Specialized drilling and well construction techniques to safely and rapidly penetrate complex, high-pressure sour zones.
3. Effective strategies to mitigate sulfur deposition in reservoirs and wellbores.
4. A complete system of corrosion control, including domestically developed corrosion-resistant materials and high-performance inhibition programs.
5. Highly efficient gas treatment processes, achieving greater than 99.9% sulfur removal in the newest facilities.
6. Stringent Health, Safety, and Environment (HSE) practices, including advanced monitoring and emergency response systems.

A defining feature of this "China model" is the emphasis on domestic innovation and manufacturing. The majority of these critical technologies—from downhole tubulars and drilling tools to gathering pipelines, desulfurization units, and catalysts—have been indigenously developed or adapted, significantly reducing capital expenditures and enhancing supply chain security.

This paper presents the first systematic overview of the technical achievements and lessons learned from the Sichuan Basin sour gas developments, updated with the latest operational data. The objective is to provide a comprehensive reference for similar projects worldwide that aim to safely and efficiently unlock sour gas reserves. The following sections detail the development history of Sichuan's sour gas fields, the key technical challenges encountered, and the suite of geological, engineering, and operational solutions that have been implemented. The discussion highlights how Chinese engineers have addressed critical issues, forming a replicable model for sour gas development that could benefit other high-H₂S provinces, such as those in the Middle East and Central Asia.

Development History of Sour Gas Fields in the Sichuan Basin

The exploitation of high-sulfur gas fields in the Sichuan Basin has evolved over more than half a century, progressing through four distinct stages of technological and operational maturity (Dou et al., 2024). The Northeast of Sichuan Basin contributes more than 80% sour gas in China, and the distribution of these sour gas fields in the Sichuan Basin is shown in Fig. 1 (Ma et al., 2008).

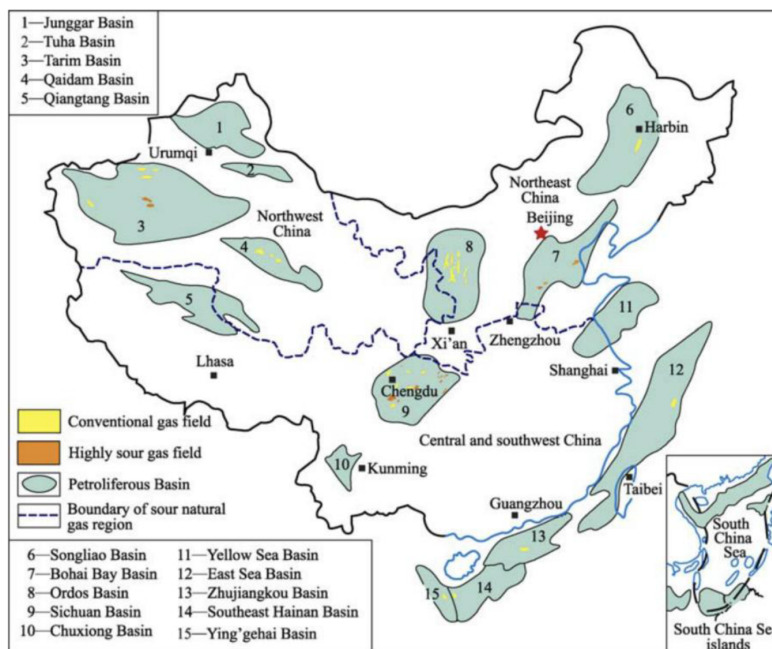


Figure 1—Distribution of sour gas fields in Sichuan basins (Ma et al., 2008)

Initial Stage (1964–1972)

China's first forays into sour gas development involved small fields such as Wei yuan (Sinian formation) and Kongtan (Lower Jialingjiang Formation). By 1972, annual sour gas production reached approximately 0.69×10^9 cubic meters, with a cumulative output of around 2.1×10^9 cubic meters. With little domestic technology to draw upon, these early operations were foundational, allowing Chinese engineers to gain essential experience in handling H_2S and laying the groundwork for future, more ambitious projects.

Steady Growth Stage (1973–2004)

This period saw the development of several medium-sized sour gas reservoirs, including the Zhongba, Wolonghe, Tiechang, and Longhuichang fields. By the end of 2004, seven high- H_2S fields were in production, and annual output had plateaued at around 2.5×10^9 cubic meters, with cumulative production reaching approximately 50.5×10^9 cubic meters. The technical focus was on the safe and moderately profitable development of these fields, which resulted in a set of mature practices for sour gas operations at this scale.

Scale-Up Stage (2005–2015)

This decade marked a transformative leap in production scale, driven by the discovery and rapid development of large, complex sour gas fields. One company brought two flagship projects online: the Puguang Triassic reef gas field, with approximately 15% H_2S , and the Yuanba Feixianguan reef gas field, the world's first ultra-deep (>7000 m) sour gas field. Concurrently, one company advanced development at fields like Longgang and Gaofengchang. By the end of 2015, Sichuan's cumulative sour gas output had surged to approximately 126.5×10^9 cubic meters. Research during this phase targeted critical technologies for large, complex reservoirs, including advanced seismic imaging, horizontal drilling, and high-efficiency gas processing, leading to the formation of a comprehensive technology suite for large-scale sour gas development. The long-term success of these projects is now evident; the Puguang field, which began production in 2009, had achieved a cumulative output of over 120 billion cube meters by 2024, while the Yuanba field had produced over 35 billion cube meters by November 2024.

Continuous Expansion Stage (2016–Present)

In recent years, development has extended to even more challenging ultra-deep and ultra-high-H₂S fields, solidifying China's technological self-sufficiency. One company, initially in partnership with Chevron, developed the Luojiashai field (first gas in 2016) before independently commissioning the Tieshanpo (TSP) field in May 2023 and the Dukouhe–Qilibei (DKH–QLB) field in June 2025. Another company has also expanded production and made significant new discoveries, including the West Sichuan field (operational March 2024), and certified major reserves at the Tongnanba and Hexingchang fields. By the end of 2021, cumulative high-sulfur gas production in the basin had reached approximately 219×10^9 cubic meters. Given the robust annual production from PetroChina Southwest (over 40 billion cube meters in 2023) and Sinopec Southwest (over 10 billion cube meters in 2024), the cumulative total has continued to rise substantially. At this stage, China has established rigorous standards and a complete, field-proven domestic technology system for safe, clean, and efficient sour gas field management, with ongoing projects representing new milestones in the industry. Throughout these stages, the Sichuan Basin sour gas industry transformed from a trial-and-error approach to a high-tech, systematically managed endeavor. The culmination of these efforts is a mature technical and management system that has ensured long-term, stable production and provided a template for future high-sulfur gas projects both domestically and internationally. Within the above four development stage, the Sichuan Basin in Southwest China boasts over 60 years of experience in sour gas development, encompassing the development of sixteen different sour gas fields and formations as shown in Table 1, making it nearly a "museum" of sour gas exploration and production.

Table 1—Main Developed High-H₂S Gas Reservoirs in the Sichuan Basin Southwest China

Sour Gas Reservoir/ Formation	Start Year	Max H ₂ S (g/cf)	Depth (ft)	Reserve classification
Weiyuan	1965.1	1586.52	10168	Large
Wolonghe	1973.08	2653.56	6546.88	Medium
Zhongba-3	1982.03	5054.4	10476.32	Medium
Wubaiti	1990.11	4186.377	9184-15416	Medium
Leiyi	1991.02	1547.91	8905.2	Large
Longhuichang-Feixian	2002.08	1937.871	11480-13776	Medium
Huanglongchang	2003.04	7072.65	13131.48	Small
Gaofengchang	2004.12	3706.56	12398.4	Small
Puguang	2006.08	7897.5	13120-19680	Largest
Longgang	2009.07	4573.53	15711.2-20368.8	Large
Damaoping	2009.09	3308.526	15252-16400	Medium
Longwangmiao	2012.12	409.968	14760-15416	Very Large
Gaomo	2012.09	772.2	16072-17712	Large
Longhuichang-changxing	2014.07	2448.225	12136-16400	Medium
Luojiashai	2016.03	7897.5	13120-19680	Large
Tieshanpo	2023.05	7987	15748	Large

Key Challenges in Sour Gas Development in Sichuan Basin

Developing sour gas fields in the Sichuan Basin and elsewhere entails a confluence of formidable technical and operational challenges, stemming from extreme reservoir conditions and the hazardous nature of the produced fluids (Zhang et al., 2016). The primary challenges include:

1. **H₂S Toxicity and Safety Risks:** Hydrogen sulfide is a highly poisonous gas; even low concentrations can be lethal, and it is flammable and explosive. Ensuring the safety of workers and surrounding communities is the paramount priority. The 2003 blowout at the LuoJia 16H well, which resulted in over 240 fatalities, serves as a stark reminder of the catastrophic potential of a sour gas release. Consequently, operations require robust H₂S detection and alarm systems, strict personal protection protocols, and comprehensive emergency response plans to manage potential leaks or blowouts. The newly operational DKH-QLB field, with an H₂S content of up to 17.06%, represents the current benchmark for the extreme toxicity levels that modern safety systems must be designed to handle.
2. **Complex Geology and Reservoir Conditions:** The sour gas fields of the Sichuan Basin are situated in complicated geological settings. These include multiple overlying pressure systems, zones prone to both lost circulation and gas kicks, and, in many cases, extensive saltwater aquifers. The reservoirs are typically deep to ultra-deep (e.g., >7,000 m in the Yuanba field), residing in naturally fractured carbonate formations characterized by significant heterogeneity and steep structural dips. These factors complicate well planning and geosteering, increasing drilling risks and demanding exceptional precision in mud weight management and wellbore stability control.
3. **Drilling and Well Construction Difficulties:** Drilling high-H₂S wells is often constrained by a "narrow safe mud weight window"—if the mud density is too low, toxic gas can influx into the wellbore; if it is too high, fragile formations may fracture, leading to costly mud losses. The simultaneous occurrence of lost circulation and gas influx is a common and dangerous problem. Other frequent issues include borehole collapse in overpressured shales, stuck pipe, and sour gas kicks. Conventional drilling and cementing methods are often insufficient, necessitating specialized technologies like managed-pressure drilling (MPD), H₂S-resistant fluid systems, and enhanced well control equipment to drill safely and efficiently.
4. **Reservoir Management under Multiphase Flow:** The productivity and ultimate recovery of sour gas reservoirs are threatened by complex multiphase flow dynamics. Many Sichuan reservoirs are subject to active edge or bottom water drive, leading to early water invasion that can kill well production and introduce additional corrosion and handling challenges. Simultaneously, as reservoir pressure and temperature decline during production, elemental sulfur can precipitate from the gas phase, clogging pore throats in the near-wellbore region and restricting flow. Predicting and managing the combined effects of this gas-liquid-solid (methane, water, sulfur) multiphase flow is a major technical challenge for long-term reservoir stewardship.
5. **Severe Corrosion:** The coexistence of H₂S, CO₂, liquid water, and sometimes elemental sulfur creates an extremely corrosive environment for downhole and surface equipment. The HPHT conditions prevalent in Sichuan's fields exacerbate corrosion rates, leading to various failure mechanisms, including uniform corrosion, pitting, sulfide stress cracking (SSC), and hydrogen-induced cracking (HIC). Standard oilfield-grade steels fail rapidly under these conditions, necessitating the use of expensive corrosion-resistant alloys (CRAs). Uncontrolled corrosion poses severe safety risks from potential sour gas leaks and can drastically shorten the operational life of critical infrastructure.
6. **High Capital and Operating Expenditures:** The development of sour gas is inherently expensive. The need for CRAs, specialized completion tools, comprehensive sulfur recovery units (SRUs), and extensive safety systems drives up capital and operating costs. The safe handling and disposal of toxic produced water and elemental sulfur also add to the operational burden. Maintaining project economics, especially as fields mature and reservoir conditions worsen (e.g., pressure depletion, increasing water production), is a persistent challenge.
7. **Stringent Environmental Protection:** Sour gas development must adhere to rigorous environmental standards. Any release of H₂S or CO₂ can harm the local environment. Furthermore, operations generate sulfur-laden wastewater and solid waste that require proper management. Modern "green"

development mandates have introduced higher requirements, such as achieving zero routine flaring and near-zero emissions of pollutants, necessitating advanced technologies for treating and reusing acid gas, contaminated water, and minimizing sulfur dioxide emissions.

In summary, high-sulfur gas fields present a nexus of difficulties: toxic gas handling, complex geology, challenging drilling conditions, multiphase production issues, aggressive corrosion, and demanding environmental and safety requirements. The subsequent sections detail the targeted technological innovations developed in China to address these multifaceted challenges (Zeng et al., 2023b).

Key Technologies for Cost-Effective and Safe Development

To address the challenges inherent in sour gas development, a suite of integrated technologies and engineering practices has been developed and implemented across the Sichuan Basin. These innovations span from subsurface characterization to surface facilities and safety systems, forming a cohesive "China solution" for sour gas. A brief summary and description of these key technologies are presented in Fig. 2, with further details on the related areas provided in the following subsections.



Figure 2—Summary of the series China solution for sour gas development in Sichuan basin (Ma et al., 2008; Zeng et al., 2023b)

Geological Characterization

The high-sulfur gas reservoirs in the Sichuan Basin are predominantly hosted in ancient marine carbonates, specifically the Late Permian Changxing and Early Triassic Feixianguan formations. These consist of reef and shoal facies limestones and dolomites, which feature complex pore systems including primary intergranular pores, secondary dissolution vugs and caverns, and extensive natural fracture networks. While reservoir zones can exhibit good porosity (5–15%) and high deliverability, the geological setting is marked by significant complexity. High heterogeneity, driven by facies variations and diagenesis, leads to irregular gas distribution. Furthermore, intense tectonic activity has resulted in steeply dipping structures (often 50–60°), complicating seismic imaging and well design.

To address this, Chinese teams have employed high-resolution seismic and "facies-controlled" modeling techniques to accurately map reservoir architecture. A "facies + seismic" constrained 3D modeling approach was developed to improve the imaging of steeply dipping reef margins, enabling precise placement of high-rate wells in identified "sweet spots." This technique was instrumental in the several sour gas fields, where contributed to all producers achieving initial production rates of over 3×10^6 m³/day (Guo et al., 2024).

A comparative analysis with other major global sour gas provinces reveals the unique geological challenges of the Sichuan Basin. The supergiant sour gas fields of the Middle East are primarily hosted in the Permo-Triassic Khuff Formation, which represents a vast, relatively stable carbonate-evaporite platform

with a "layer-cake" geometry. While also a complex carbonate system, the Khuff's depositional setting was a low-relief shelf, leading to more laterally continuous reservoir units compared to the often isolated reef buildups found in Sichuan. The Feixianguan and Changxing formations in Sichuan were subject to more intense tectonic deformation, resulting in greater structural complexity and compartmentalization. This geological distinction underscores why advanced, high-resolution characterization is not just beneficial but essential for successful development in the Sichuan Basin, as it forms the basis for all subsequent engineering decisions.

Reservoir Management

Managing sour gas reservoirs requires meticulous control of production processes to mitigate productivity decline from sulfur precipitation and water influx, two phenomena that can severely impact ultimate recovery.

Sulfur Deposition Management. As high- H_2S gas is produced, reductions in pressure and temperature cause dissolved elemental sulfur to precipitate, clogging pore throats and wellbores. In the Puguang field, where the average reservoir temperature is 128°C , sulfur precipitates in a liquid form, which exhibits a strong tendency to absorb in and plug smaller pore spaces. This can cause severe permeability damage, with reductions exceeding 90% in reservoirs with poorer initial properties, and up to 70–80% in the near-wellbore region of productive wells. The gas-water-sulfur three-phase wellbore deposition prediction model considering the influence of diffusion and adsorption results are shown in Fig.3 (Zeng et al., 2024).

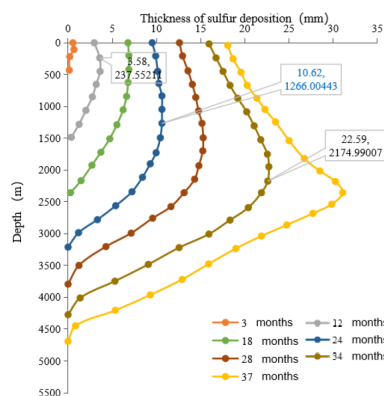


Figure 3—Diagram of Sulfur deposition location and thickness in wellbore (Zeng et al., 2024)

To manage this threat, a multi-pronged strategy has been implemented:

1. **Prediction and Modeling:** Operators utilize a combination of geochemical thermodynamic models to predict the pressure-temperature-composition conditions for sulfur saturation, kinetic models to estimate the rate of deposition, and increasingly, data-driven machine learning algorithms to provide real-time risk assessments based on historical production data. These predictive tools are crucial for anticipating where and when sulfur plugging will occur, allowing for proactive intervention (Hu et al., 2024).
2. **Mitigation and Remediation:** A suite of interventions has been deployed. Proactive measures include the injection of chemical inhibitors or scavengers to prevent sulfur from forming. Remedial actions are critical for restoring productivity in plugged wells. Field experience in Sichuan has demonstrated that acid fracturing is a particularly effective remediation technique. Unlike sulfur solvents, which show limited penetration, acid treatments not only dissolve solid sulfur but also etch new flow channels in the carbonate formation, bypassing the damaged zone and providing more durable productivity

restoration. This finding has become a best practice, and its application in several fields has been key to sustaining high production rates (Xiong et al., 2024).

Water Invasion Management. Many of Sichuan's sour gas reservoirs are connected to active aquifers, making water invasion a primary long-term production challenge. Water influx reduces gas relative permeability, accelerates corrosion, and complicates surface handling. In these highly fractured carbonate reservoirs, water can channel rapidly through high-permeability fractures, bypassing large volumes of gas in the matrix and leading to premature well death. The diagram of Water invasion of the sour gas field in Sichuan Basin is shown in Fig.4 (Zeng et al., 2023a).

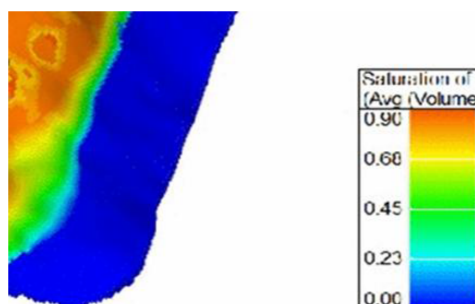


Figure 4—Diagram of Water invasion of the sour gas field in Sichuan Basin

To counter this, Chinese engineers have developed a sophisticated conformance control strategy known as "upper choke, lower drain & block". This differentiated production approach involves:

1. **Reservoir Monitoring and Modeling:** The strategy is guided by detailed reservoir surveillance, including high-resolution simulation models to predict water front movement and time-lapse (4D) seismic surveys to visualize saturation changes over time.
2. **Upper Choke:** Wells located in structurally high parts of the reservoir (crestal wells) are produced at intentionally restricted rates. This "choking" minimizes localized pressure drawdown, slowing the vertical rise of the gas-water contact and delaying water conning.
3. **Lower Drain & Block:** Wells in lower structural positions, which are more susceptible to water breakthrough, are managed aggressively. This includes performing water shutoff treatments, where gels or resins are injected to "block" water-producing fractures. In some cases, wells are equipped with downhole pumps or velocity strings to continuously remove encroaching water, effectively converting them into "drainage" wells that help manage the aquifer energy across the field.

This coordinated strategy promotes a more uniform advancement of the water front, preventing premature flooding of high-productivity wells and maximizing the swept area, thereby enhancing the ultimate recovery factor. In the Puguang field, this approach has proven effective at controlling water invasion and maintaining stable production. As fields mature and reservoir pressure declines, this strategy is complemented by the timely installation of surface compression (boosting) to lower wellhead back-pressure and extend the productive life of wells.

Drilling and Completion

Drilling and completing wells safely and efficiently in the deep, HPHT, and high-H₂S environment of the Sichuan Basin represents a major technical achievement. Innovations in drilling fluids, tools, and operational procedures have dramatically reduced drilling times while enhancing safety (Zhang et al., 2023a). The diagram of well integrity design with material selection is illustrated in Fig.5.

1. **High-Angle Directional Drilling and ROP Improvement:** To navigate the structurally complex formations with steep dips, specialized directional drilling techniques combining vertical drilling systems and rotary steerable assemblies have been developed to maintain trajectory control. Regional rock mechanical studies and the creation of a drill bit performance database have led to the custom design of advanced polycrystalline diamond compact (PDC) bits, significantly increasing the rate of penetration (ROP). The application of continuous measurement-while-drilling systems has enabled "one-trip" drilling through key intervals, further reducing drilling days (Jia et al., 2023).
2. **Managed-Pressure Drilling (MPD) and Advanced Cementing:** The narrow pressure window is one of the most critical drilling challenges. Managed-pressure drilling (MPD) technology has been widely adopted to provide precise, real-time control of bottom-hole pressure, minimizing the simultaneous risks of gas kicks and fluid losses. Securing well integrity through effective cementing is equally vital. A novel low-density, tough, gas-blocking cement slurry system, combined with "precise controlled-pressure balanced cementing" techniques, has enabled the successful isolation of high-H₂S zones, achieving high-quality cement bonds even in severe loss-circulation environments (Zheng et al., 2023b).
3. **Enhanced Well Control and Completion:** Given the lethal hazard of H₂S, well control is paramount. An "evaluation–design–control integrated well control technique" has been formulated, supported by upgraded blowout preventer (BOP) stacks with H₂S-rated components. A key challenge in well control for high-H₂S kicks is the complex physicochemical behavior of the gas. H₂S is significantly more soluble in drilling fluids than methane and can undergo phase transitions within the wellbore. As a kick is circulated to the surface, the rapid release of dissolved gas and phase changes can cause sudden volume expansion and a sharp drop in annular pressure, inducing a secondary, more dangerous influx. To manage this, operators rely on advanced transient two-phase flow models that incorporate gas solubility and phase behavior to predict the required wellhead back-pressure for safe circulation. Completions are also designed for safety, using a two-step process to manage initial flow under maximum control and installing permanent downhole pressure gauges for continuous, life-of-well monitoring. These integrated drilling and completion technologies have been instrumental in enabling the safe and efficient construction of even the most challenging ultra-deep, ultra-sour wells, with the expertise now fully localized in China (Yuan et al., 2024).

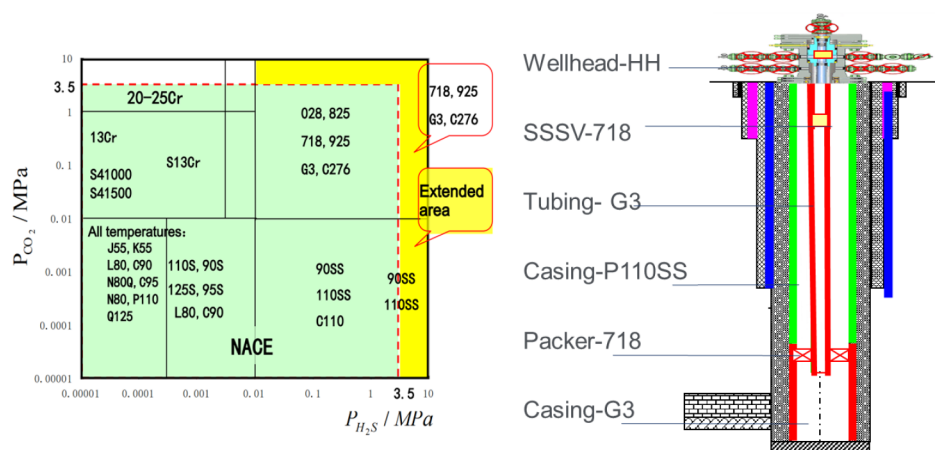


Figure 5—Diagram of well integrity design (Zhang et al., 2023a)

Production Operations and Field Management

Field operations in Sichuan's sour gas projects are governed by a "zero harm" philosophy, with extensive systems in place to prevent incidents.

1. **Stringent H₂S Safety Systems:** Facilities are designed with strict safety exclusion distances to protect public communities. Multi-level H₂S detection and alarm systems are installed throughout, with automatic triggers for emergency shutdown (ESD) and blowdown systems that can rapidly isolate and depressurize equipment to limit the scale of any potential release.
2. **Remote and Automated Operations:** To minimize human exposure, operations are highly automated. Wells and gathering stations are monitored and controlled remotely from a central control room. This "digitally guided, minimal manning" strategy is a core tenet of modern sour gas field management.
3. **Pipeline Integrity and Leak Detection:** The sour gas gathering pipeline network, a critical risk area, is equipped with advanced monitoring systems. The Chuandongbei network utilizes fiber-optic distributed sensing technology, which can detect the acoustic and thermal signatures of a gas leak in real-time and pinpoint its location, enabling immediate response. This technology was implemented as a direct lesson from the tragic 2003 gas release incident and has contributed to an exemplary safety record for the pipeline network in recent years.
4. **Emergency Preparedness and Environmental Control:** Rigorous emergency drills are standard practice. The deliberate six-month delay in starting up the Tieshanpo field to ensure all safety systems were fully tested exemplifies the prevailing safety-first culture. Operationally, a commitment to "zero routine flaring and zero emissions" is maintained through technologies like flash vapor compression and recovery systems for produced water and comprehensive wastewater treatment, ensuring that sour gas fields are operated cleanly and sustainably.

Corrosion Control

Corrosion control is a critical enabler for the long-term integrity and safety of high-sulfur gas infrastructure (Han et al., 2011). The aggressive environment created by high partial pressures of H₂S and CO₂ in the presence of water requires a comprehensive, lifecycle approach to corrosion management. China has achieved self-sufficiency in this domain, localizing all key corrosion control technologies (Yan et al., 2024).

1. **Material Selection and Analysis:** A rigorous methodology for material evaluation was established, supported by a unique field-exposure corrosion test loop that simulates actual production conditions. This has led to the creation of a material performance database and the large-scale domestic manufacturing and application of bimetal composite pipes and sour-resistant alloys, replacing expensive imports and ensuring infrastructure is built with materials proven to withstand the specific corrosive environment. The extended NACE chart for material selection is displayed in Fig.6.
2. **High-Performance Corrosion Inhibitors:** Chinese research teams have independently developed and industrialized new corrosion inhibitor formulations tailored for high-sulfur service. These include long-lasting film-forming inhibitors, innovative foam-suppressing inhibitors that solve operational issues in gathering systems, and anti-freeze inhibitors for cold spots. These domestically produced chemicals have shown superior performance compared to international products and have significantly extended the service life of carbon steel infrastructure. Fig.7 illustrated the workflow of corrosion inhibitor injection.
3. **Advanced Corrosion Monitoring and Detection:** Management has shifted from reactive to proactive through the implementation of a remote online corrosion monitoring system that covers the entire production network in real-time. This system integrates data from multiple sources (corrosion coupons, electrical resistance probes, ultrasonic sensors) to create quantitative "corrosion maps" that identify high-risk hotspots. This data-driven approach supports a risk-based inspection (RBI) regime, optimizing maintenance resources while ensuring no severe corrosion goes undetected. Fig.8 shows the general distribution of monitor point in surface facility.
4. **Intelligent Corrosion Management:** All data streams are integrated into an intelligent corrosion diagnosis and control platform. This digital system uses big data and AI algorithms for trend analysis,

intelligent anomaly identification, and real-time early warnings. It provides decision support by recommending optimal mitigation actions, transforming corrosion management from periodic manual checks to a predictive, digitally-enabled maintenance model.

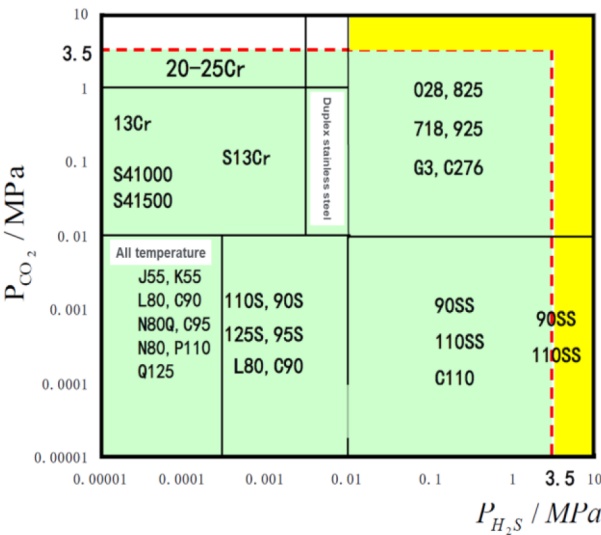
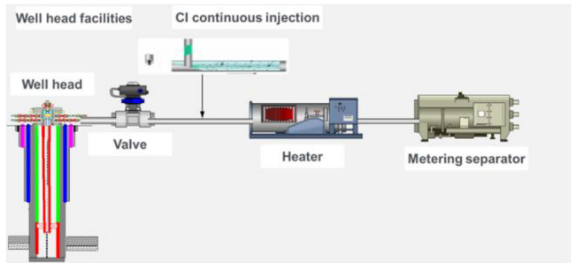


Figure 6—The extended NACE chart for material selection

CI INJECTION TECHNOLOGY AND PROCESS IN WELLHEAD:



CI INJECTION TECHNOLOGY AND PROCESS IN PIPELINE:

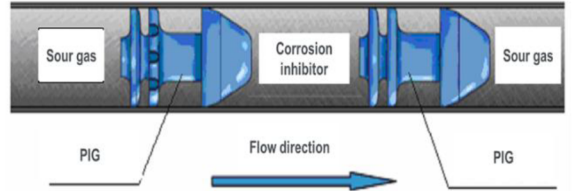


Figure 7—Workflow of corrosion inhibitor injection

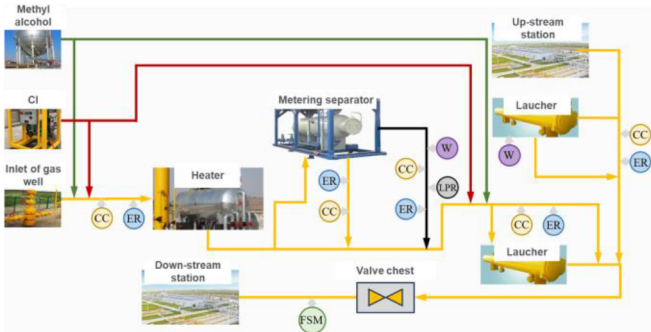


Figure 8—Distribution of monitor point in surface facility

Gas Purification and Sulfur Recovery

Gas purification is the centerpiece of surface processing, ensuring that the final product meets sales gas specifications while capturing elemental sulfur and minimizing environmental impact. China has deployed entirely domestic process technology and catalysts in its largest sour gas plants, achieving world-class performance (Xiong et al., 2023).

1. **Gas Sweetening:** The produced gas is first treated in an amine sweetening unit. MDEA-based alkanolamine processes have been optimized with proprietary solvent formulations that selectively absorb H_2S over CO_2 . This selectivity reduces the required solvent circulation rate and reboiler energy consumption, lowering operating costs.
2. **Sulfur Recovery and Tail Gas Treatment:** The acid gas from the amine unit is fed to a Sulfur Recovery Unit (SRU), which uses the Claus process to convert H_2S into elemental sulfur. A major achievement in the Sichuan plants is the attainment of ultra-high sulfur recovery efficiencies, often exceeding 99.8%. This is accomplished by combining domestically developed, high-performance Claus catalysts with a comprehensive Tail Gas Treatment Unit (TGTU). Advanced TGTU processes, such as the Shell Claus Off-gas Treating (SCOT) process, are implemented to treat the residual sulfur compounds in the Claus tail gas. In the SCOT process, all sulfur species in the tail gas are hydrogenated back to H_2S , which is then selectively absorbed by an amine solvent and recycled to the front of the Claus plant, ensuring near-complete sulfur capture and minimal SO_2 emissions. The newest facilities, like West Sichuan Gas Plant, feature closed-loop systems with over 99.9% sulfur recovery, zero wastewater discharge, and zero routine flaring.
3. **Sulfur Handling:** The millions of tons of elemental sulfur produced annually are a valuable industrial commodity. Modern plants feature safe handling systems, including molten sulfur degassing and granulation units that produce sulfur pastilles or pellets for easy and safe transport, turning a hazardous waste stream into a revenue source that improves project economics.

Monitoring and Intelligent Field Management

Modern sour gas operations in Sichuan are underpinned by extensive monitoring and digitalization, transforming them into "smart fields" that leverage a digital twin approach to enhance safety and efficiency. This evolution from basic digitalization to fully integrated intelligent operations is a key feature of the modern "China Model."

The development of such complex and hazardous fields is increasingly reliant on these advanced digital platforms. The extreme toxicity and corrosivity of ultra-sour gas create an operational environment with a very low tolerance for error, where a minor process upset could have severe consequences. Traditional management methods based on periodic inspections and reactive problem-solving are insufficient to manage this level of risk. The "Intelligent Field" or "Digital Twin" paradigm fundamentally alters this by shifting operations from a reactive to a predictive stance. It is this layer of predictive, real-time, integrated oversight that reduces operational risk to an acceptable level, making the development of fields like Tieshanpo and Dukouhe-Qilibei technically and economically feasible. Thus, digitalization is not merely an optimization tool but a critical enabling technology for these frontier projects.

1. **Architecture of an Intelligent Gas Field:** The Puguang gas field serves as a prime case study for this transition, with an architecture designed around "one platform, two centers, and two systems". This framework consists of an integrated application platform, a resource sharing center (for cloud computing and big data), an intelligent assistant decision-making center (using AI and machine learning), and overarching systems for standards and information security. The intelligent field management of Puguang sour gas field is shown in Fig.9.

2. **Digital Twin Applications:** This architecture is the foundation for a comprehensive digital twin—a virtual replica of the entire production system, from the reservoir to the processing plant, continuously updated with real-time data from a network of IoT sensors. The initiative is driving the implementation of this technology across its operations. This virtual model enables a range of high-value applications:
 - a. **Predictive Maintenance:** AI algorithms analyze real-time data from equipment sensors (e.g., vibration, temperature) to predict potential failures, allowing maintenance to be scheduled proactively, thus minimizing downtime and preventing catastrophic breakdowns.
 - b. **Production Optimization:** The digital twin integrates reservoir, wellbore, and surface facility models to simulate different operational scenarios. This allows operators to optimize production allocation, manage pressure, and adjust strategies to prevent issues like sulfur deposition or water conning before they impact output.
 - c. **Enhanced Safety and Remote Operations:** The platform provides a "single pane of glass" view of the entire field's health, integrating data from H₂S sensors, optical gas imaging cameras, and pipeline monitoring systems. This enhances situational awareness while allowing for remote control of operations, significantly reducing personnel exposure to hazardous environments.
 - d. **Emergency Response Simulation:** The digital twin can be used to simulate emergency scenarios, such as gas leaks or blowouts, providing a safe and realistic environment for training personnel and validating emergency response plans.

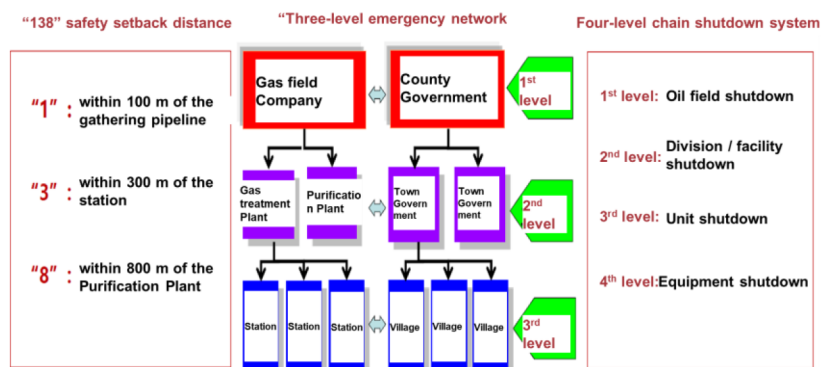


Figure 9—The intelligent field management of Puguang sour gas field (Zeng et al., 2023b)

The investment in this digital infrastructure has proven to be a critical factor in ensuring that Sichuan's sour gas fields operate not only at peak efficiency but also with the highest levels of safety, making real-time insight paramount to success in this high-risk environment.

Global Context and Comparative Analysis

The technical achievements in the Sichuan Basin are best understood when placed in the context of other major sour gas developments worldwide. This comparison highlights the unique characteristics and strategic drivers of the "China Model."

Comparison with Middle Eastern Sour Gas Fields

The Middle East holds some of the world's largest sour gas reserves, and its development approach is characterized by capital-intensive megaprojects.

Case Study: ADNOC's Shah Gas Field (UAE): The Shah field is a global benchmark, often cited as the world's largest ultra-sour gas operation. It processes approximately 1.28 billion standard cubic feet per day (bscfd) of raw gas with a very high H₂S content of 23% and produces 4.2 million tons of sulfur per year. The project required a \$10 billion investment and was developed as a joint venture with an international

oil company (Occidental Petroleum) to leverage specialized expertise. Ongoing expansion projects aim to further increase this capacity. The development model for Shah and other regional projects, like Hail and Ghasha, is defined by immense scale, high capital investment, and strategic partnerships with IOCs.

Comparison with Central Asian Sour Gas Fields

Central Asia's supergiant fields represent an extreme frontier of operational difficulty, where environmental and subsurface challenges converge.

Case Study: Kashagan & Tengiz (Kazakhstan): These fields are notorious for their complexity. The Kashagan field contains gas with ~17-19% H₂S at extremely high pressure, located under the shallow, icy waters of the North Caspian Sea, which experiences extreme temperature variations. Its development has been one of the most expensive in history (>\$50 billion) and has faced severe technical setbacks, most notably the catastrophic failure of its initial pipelines due to sulfide stress cracking, which necessitated a complete replacement with highly expensive inconel-cladded pipes. The Tengiz field also features very high H₂S content and pressure, requiring a massive sour gas re-injection program for pressure maintenance and facing logistical challenges in managing vast stockpiles of produced sulfur. These projects required bespoke, often unprecedented engineering solutions to overcome a combination of extreme subsurface and surface conditions.

Distinguishing Features of the "China Model"

The comparative analysis reveals that the Sichuan development strategy, or the "China Model," is distinct from the megaproject and frontier-engineering models seen elsewhere. It is characterized by a different set of strategic priorities and technological pathways.

1. **Rapid Technology Localization and Cost Control:** A core tenet of the Sichuan model is a deliberate, systematic strategy to move from importing technology and expertise to developing a complete domestic supply chain for critical sour service equipment and consumables. This includes corrosion-resistant alloys, high-performance inhibitors, specialized catalysts, and advanced drilling tools. This focus on localization is explicitly aimed at reducing capital and operating expenditures, making a portfolio of challenging projects economically viable without relying on the massive budgets or risk profiles typical of IOC-led megaprojects.
2. **Systematic, Portfolio-Based Development:** Rather than focusing on a single, massive project, the Sichuan development is a continuous program of developing a portfolio of fields. This approach allows for rapid learning and iteration, where technologies and operational lessons from one project (e.g., Puguang) are refined and quickly applied to the next, more challenging one (e.g., Tieshanpo). This creates a powerful feedback loop that accelerates technological maturation and de-risks subsequent developments.
3. **Integrated Digital-First Approach as an Enabler:** For the newest ultra-sour fields, intelligent platforms and digital twins are not an afterthought for optimization but a foundational component of the project design. This digital-first approach is a strategic necessity to manage the exceptionally high operational risks associated with ultra-high H₂S concentrations. By enabling predictive maintenance, real-time risk monitoring, and remote operations, these digital systems reduce the probability of catastrophic failure to a level that makes these frontier projects insurable and operable, effectively enabling their development.

Conclusion

The successful large-scale development of sour gas fields in the Sichuan Basin stands as a significant achievement in the global energy industry, demonstrating how concerted research, targeted engineering innovation, and systematic management can overcome formidable technical and safety challenges. Over

the past decade, China has refined a comprehensive set of "China solutions" that have enabled the safe, efficient, and economically viable exploitation of some of the world's most challenging high-H₂S, high-CO₂ gas reservoirs. The "China Solution" represents a comprehensive framework that addresses every challenge from subsurface uncertainty to surface emissions, driven by a philosophy of autonomous innovation, proactive risk engineering, and dynamic asset management.

The knowledge and experience gained from the Sichuan Basin offer a valuable and distinct blueprint for other high-sulfur gas provinces worldwide. In contrast to capital-intensive megaprojects or high-cost frontier engineering solutions seen elsewhere, the "China Model" is characterized by its focus on cost-effectiveness, rapid and iterative technological development across a portfolio of projects, and the foundational integration of digital twin-based intelligent field management as a key enabler for managing operational risk. As the global energy industry continues to seek diverse and cleaner fuel sources, the ability to safely and economically harness sour gas expands the available natural gas supply. The advancements made in China's sour gas field development are not only a national strategic asset but also a significant contribution to the global knowledge base, paving the way for the next generation of sour gas projects to be safer, greener, and more efficient than ever before.

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